

CM0081 Formal Languages and Automata

§ 9.3 Undecidable Problems About Turing Machines

Andrés Sicard-Ramírez

Universidad EAFIT

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Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

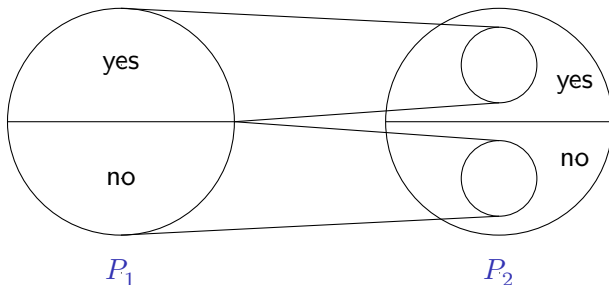
Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Reductions

Definition

Let P_1 and P_2 be two problems. A **reduction** from P_1 to P_2 is a **Turing machine** that takes an instance of P_1 written on its tape and **halts** with an instance of P_2 that have the **same** answer (i.e. a reduction is an algorithm).



Reductions

Theorem 9.7

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If there is a reduction from P_1 to P_2 then:

- (i) if P_1 is undecidable then so P_2 ,
- (ii) if P_1 is not recursively enumerable then so P_2 .

Proof

Hint: Suppose the P_2 is decidable/recursively enumerable and find a contradiction.

Turing Machines that Accept the Empty Language

Notation

Henceforth, we'll regard strings as the Turing machines they represent.

Two languages

Let $\Sigma = \{0, 1\}$. Then

$$\begin{aligned}L_e &:= \{ M \in \Sigma^* \mid L(M) = \emptyset \}, \\L_{ne} &:= \{ M \in \Sigma^* \mid L(M) \neq \emptyset \}.\end{aligned}$$

Turing Machines that Accept the Empty Language

Theorem 9.8

The language L_{ne} is recursively enumerable.

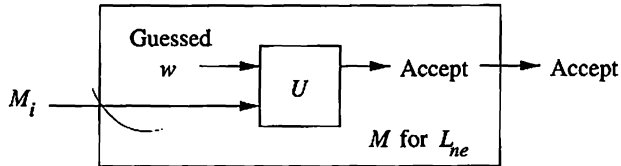
Turing Machines that Accept the Empty Language

Theorem 9.8

The language L_{ne} is recursively enumerable.

Proof

Construction of a non-determinist Turing machine to accept L_{ne} (Fig. 9.8):



Turing Machines that Accept the Empty Language

Theorem 9.9

The language L_{ne} is not recursive.

Turing Machines that Accept the Empty Language

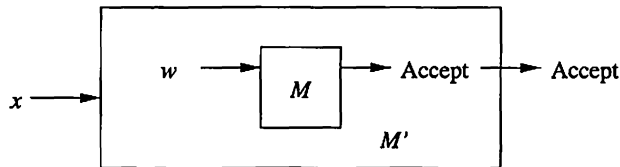
Proof

1. Reduction from L_u to L_{ne} where the pair (M, w) is converted in M' , such that $w \in L(M)$ iff $L(M') \neq \emptyset$.

Turing Machines that Accept the Empty Language

Proof

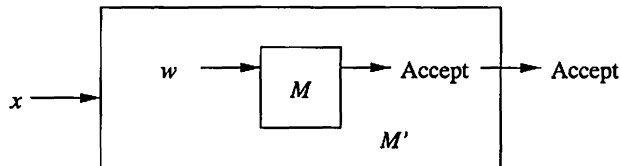
1. Reduction from L_u to L_{ne} where the pair (M, w) is converted in M' , such that $w \in L(M)$ iff $L(M') \neq \emptyset$.
2. The key is that M' ignores its input (Fig. 9.9).



Turing Machines that Accept the Empty Language

Proof

1. Reduction from L_u to L_{ne} where the pair (M, w) is converted in M' , such that $w \in L(M)$ iff $L(M') \neq \emptyset$.
2. The key is that M' ignores its input (Fig. 9.9).



3. L_{ne} is not recursive by Theorem 9.7. ■

Turing Machines that Accept the Empty Language

Theorem 9.10

The language L_e is not recursively enumerable.

Turing Machines that Accept the Empty Language

Theorem 9.10

The language L_e is not recursively enumerable.

Proof

Hint: The language L_e is the complement of the language L_{ne} .

Rice's Theorem

Definition

The **set of the recursively enumerable languages**, denoted **RE**, is the set

$$\text{RE} := \{ L \subseteq \Sigma^* \mid L \text{ is a recursively enumerable language} \}.$$

Rice's Theorem

Definition

A **property** P of the recursively enumerable languages is a subset of RE , that is, $P \subseteq \text{RE}$.

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- $P(L)$: L is a language regular.

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- $P(L)$: L is a language regular.
- $P(L)$: L is finite.

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Definition

A **property** P of the recursively enumerable languages is a subset of RE , that is, $P \subseteq \text{RE}$.

Example

- $P(L)$: L is a language regular.
- $P(L)$: L is finite.
- Trivial properties: $P(L) = \emptyset$ or $P(L) = \text{RE}$.

Rice's Theorem

Theorem 9.11 (Rice's theorem, first version)

Every non-trivial property of **RE** is undecidable (Rice 1953).

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How to prove Rice's theorem?

We identify a property P by the Turing machines M such that $L(M) \in P$.

Rice's Theorem

Theorem 9.11 (Rice's theorem, first version)

Every non-trivial property of $\mathbf{RE}$ is undecidable (Rice 1953).

How to prove Rice's theorem?

We identify a property P by the Turing machines M such that $L(M) \in P$.

Theorem (Rice's theorem, second version)

If $P \subseteq \mathbf{RE}$ is a non-trivial property then

$$L_P := \{ M \in \Sigma^* \mid L(M) \in P \}$$

is undecidable.

Rice's Theorem

Proof

Case $\emptyset \notin P$.

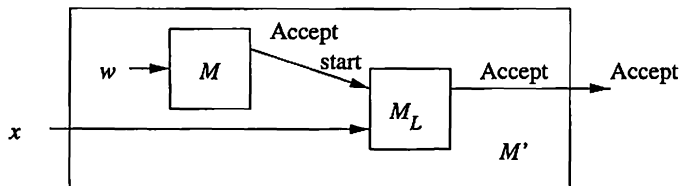
1. Let L be a language and M_L be a Turing machine such $L \neq \emptyset$, $L \in P$ and $L = L(M_L)$. Reduction from L_u to L_P where the pair (M, w) is converted in M' such that (Fig. 9.10):
 - (i) $L(M') = \emptyset$ (i.e. $M' \notin L_P$) if $w \notin L(M)$ and
 - (ii) $L(M') = L$ (i.e. $M' \in L_P$) if $w \in L(M)$.

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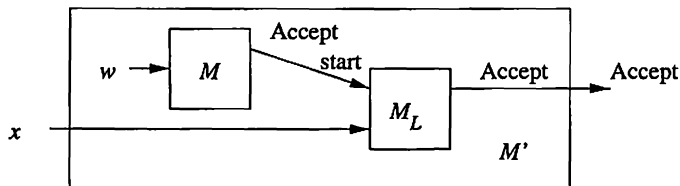


Rice's Theorem

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 - $L(M') = \emptyset$ (i.e. $M' \notin L_P$) if $w \notin L(M)$ and
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- L_P is not recursive by Theorem 9.7.a.

Rice's Theorem

Proof (continuation)

Case $\emptyset \in P$.

1. By the previous case, $\overline{P}$ is undecidable, i.e. $L_{\overline{P}}$ is undecidable.
2. $\overline{L_P} = L_{\overline{P}}$.
3. Suppose L_P is decidable then $\overline{L_P}$ would be also decidable (contradiction).
4. Therefore, L_P is undecidable.



Rice's Theorem

Remark

All problems about Turing machines that involve **only the language** that the Turing machine accepts are undecidable.

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Examples

- Is the language accepted by the Turing machine empty? Is it finite? Is it regular? Is it context-free?
- Does the language accepted by the Turing machine contain the string “hello world”? Does it contain all the even numbers?

Rice's Theorem

Remark

Rice's theorem does not imply that **everything** about Turing machines is undecidable.

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Example

It is decidable if a Turing machine has five states.

References

- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
- H. G. Rice (1953). Classes of Recursively Enumerable Sets and Their Decision Problems. Transactions of the American Mathematical Society 74.2, pp. 358–366. DOI: [10.1090/S0002-9947-1953-0053041-6](https://doi.org/10.1090/S0002-9947-1953-0053041-6) (cit. on pp. 21–23).